

Superrdev Patch Exercise

Bug 1 → Search shows wrong tasks

Where:

- Backend → TaskRepository.java (search query)
- Same bug in db/queries/search_tasks.sql
- Same bug in db/oracle/task_search_package.sql

How found:

- Status = Done + empty search → still showed Open & In Progress
- Searched "legacy" → hidden (archived) task showed up

Root cause:

- SQL does AND before OR
- Query had no brackets
- So status check + archived check only worked on half the query

Fix:

- Added brackets around title & description check
- Now → not archived AND (title OR description matches) AND status matches
- Fixed all 3 files
- Also sort by id → page order stays same

Bug 2 → Search slow for no reason

Where:

- Backend → TaskController.java

How found:

- Short searches took approx. 1 sec
- Found Thread.sleep in the code

Root cause:

- Code paused on purpose
- Shorter text = longer pause
- Did no real work

Fix:

- Removed the pause
- Search now ~15 ms

Bug 3 → Page stuck on Loading

Where:

- Frontend → useTasks.js

How found:

- Stopped server + changed filter
- Page showed Loading forever

Root cause:

- On error, loading never turned off
- Screen checks loading first → error hidden
- Old error never cleared

Fix:

- Clear error before each request
- Always turn loading off at the end (finally)
- Now error message shows

Bug 4 → Old results replace new ones

Where:

- Frontend → useTasks.js + api.js

How found:

- Typing a, ap, api → sends 3 requests
- Answers can come back in any order
- Answer for "a" can come last & replace "api"

Root cause:

- Old requests never cancelled
- Every answer saved, even old ones

Fix:

- Cancel old request when search changes (AbortController)
- Wait 300 ms after typing stops before searching
- Now only newest answer shows + fewer requests sent

Bug 5 → Stuck on empty page

Where:

- Frontend → App.jsx

How found:

- Went to page 4, then typed a search
- Showed "No tasks found" + no page buttons

Root cause:

- Page number stayed at 4
- New search only had 1 page

Fix:

- Go back to page 1 when search or status changes

Bug 6 → Server crashes on bad input

Where:

- Backend → TaskController.java

How found:

- status=foo → 500 error
- page=0 → 500 error
- Very big page number → also crashed

Root cause:

- Wrong status throws an error
- Page 0 or huge page → negative number → list cut crashes
- Input never checked

Fix:

- Check input first
- Return 400 with clear message
- pageSize must be 1 to 100
- Use bigger number type (long) so it can't go negative

Small fixes

- % and _ now searched as normal letters
- Logger instead of System.out.println
- Removed console.log from frontend
- Page size 10 → now one constant
- Screen shows the server error message

How I tested

- Ran backend + frontend with README commands
- Tried API links: status, search, wrong status, page 0, big page, - and %
- Browser: page 4, then search → went back to page 1
- Stopped server → page showed an error, not Loading